

DoingLanguage

Agentic Speech & Language Rehabilitation Ecosystem

DEVPOST HACKATHON PROJECT

\$50,000 GRANT TRACK

PATIENT & SLP SYNERGY

OFFLINE-FIRST PWA

PRINCIPLES OF MOTOR LEARNING

1. Executive Summary & Elevator Pitches

Micro-Pitch (< 200 Characters)

DoingLanguage is an accessible, offline-ready PWA helping people with apraxia and aphasia rebuild speech with interactive speech recognition and progressive, evidence-based motor learning exercises.

Exact Length: 198 characters (including spaces & punctuation)

Short Pitch (< 150 Words)

DoingLanguage is a rehabilitation PWA designed for individuals recovering from stroke, brain injury, apraxia, or aphasia. Rebuilding communication requires high-repetition practice extending beyond clinic walls. Grounded in clinical Principles of Motor Learning, its 10-tier curriculum systematically advances users from phonemes to fluent conversation with real-time speech recognition, voice synthesis, and dynamic multi-modal cueing. A patience-first design eliminates timers and penalties, empowering learners to practice at their own pace and reclaim their voice.

Word Count: 79 words

2. Inspiration & The Lived Experience

The Clinical Context

In May 2024, Richard sustained a left middle cerebral artery (MCA) embolic stroke alongside comorbid hypertension, hyperlipidaemia, and recent localized prostate cancer surgery. While retaining independence in personal care and mobile without assistance, post-operative fatigue slowed progress, leaving residual dysphasia, reduced right-hand fine dexterity, and mild right lower-limb weakness.

Working to optimize speech therapy through Australia's National Disability Insurance Scheme (NDIS) highlighted a critical systemic challenge: **The Therapy Cliff**. While clinical Speech-Language Pathologists (SLPs) provide expert guidance, appointments are typically 1 hour per week. Yet neuroplasticity and the **Principles of Motor Learning (PML)** mandate high-volume, repetitive practice:

Optimal Neuroplastic Stimulus: **$R \geq 50$ to 100** active productions per target per day

DoingLanguage was created as an agentic companion linking the Patient and the SLP. The patient receives an empathetic, fatigue-aware, patience-first practice environment, while the SLP receives objective acoustic and phonetic telemetry to calibrate prescriptions.

3. The 10-Tier Clinical Curriculum

Tier	Domain	Target Focus	Interaction Model
Tier 1	Foundations	Letters (A-Z), Numbers (0-100), 44 English Phonemes	See & Hear, Match, Name
Tier 2	Short Words	High-frequency 2-3 letter words, Consonant-Vowel-Consonant	Phoneme Isolation, STT
Tier 3	4-Letter Words	Initial & final consonant blends, silent-e rules	Spoken Production & Cues
Tier 4	Compound Words	Multi-syllabic segmentation (sun-flower, bed-room)	Syllable Pacing & Audio
Tier 5	Everyday Objects	Categorized vocabulary (food, household, clothing)	Picture Naming & Recall
Tier 6	Actions & Verbs	High-utility verbs and present-continuous tense	Sentence Completion
Tier 7	Carrier Phrases	2-4 word functional phrases ("I want...", "Where is...")	Rhythmic Speech Pacing
Tier 8	Full Sentences	Subject-Verb-Object structures and daily requests	Speech Recognition Scoring
Tier 9	Functional Dialogue	Doctor visits, shopping, emergency interactions	Turn-taking Agent Prompts
Tier 10	Discourse & Stories	Paragraph reading, picture description, open expression	Narrative Evaluation

4. UML Specification & System Diagrams

Figure 1: UML Use Case Diagram

BEHAVIORAL MODEL

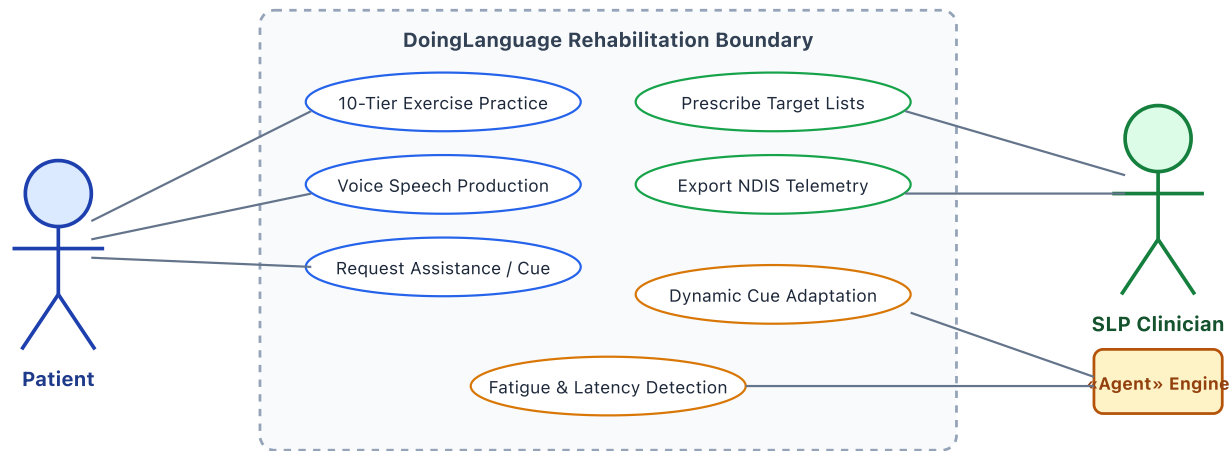


Figure 2: UML Component Architecture Diagram

STRUCTURAL MODEL

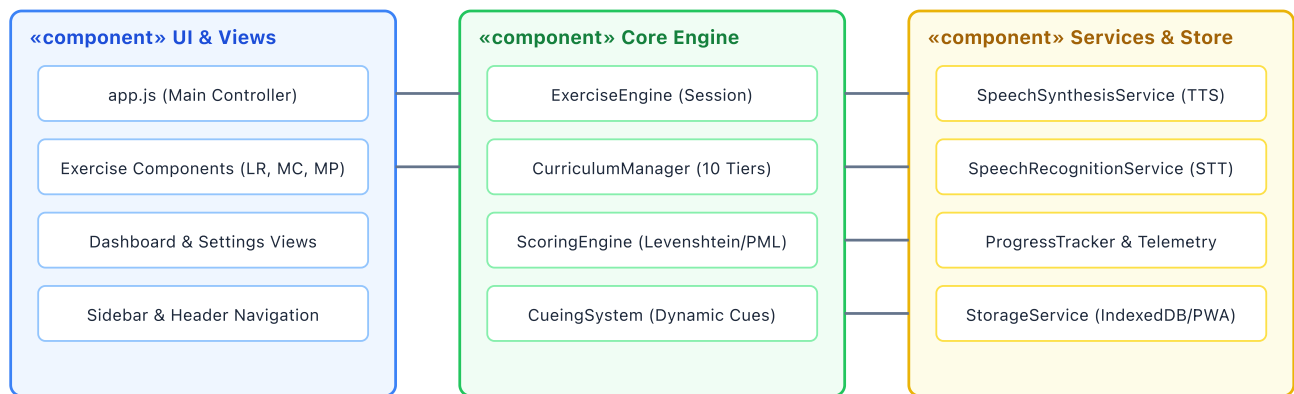


Figure 3: UML Class Diagram

STATIC DESIGN

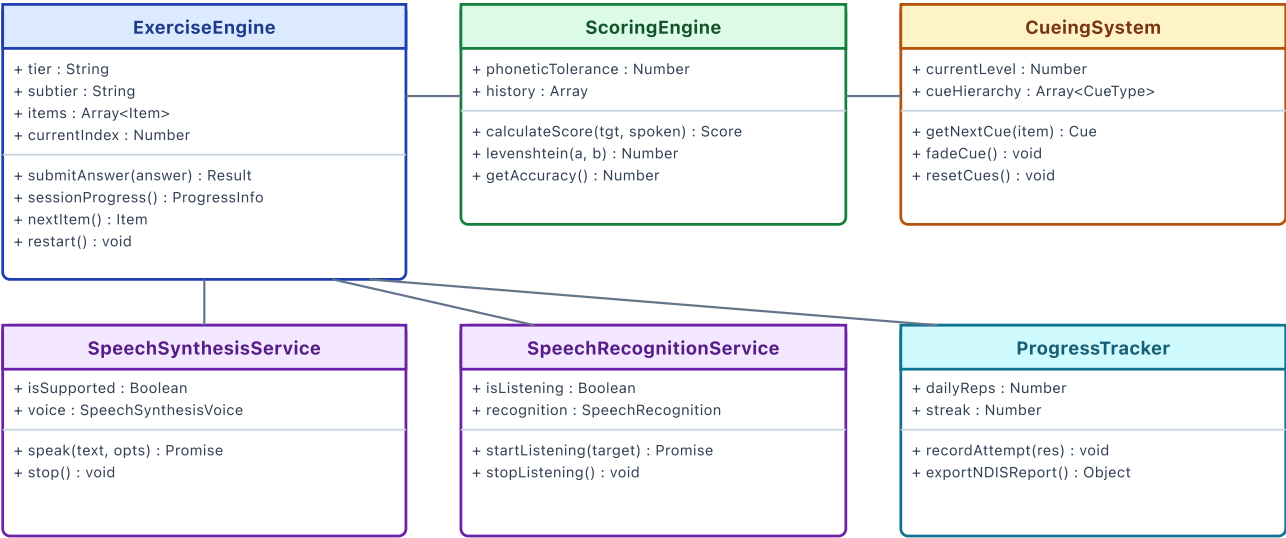


Figure 4: UML Sequence Diagram (Speech Loop)

INTERACTION FLOW

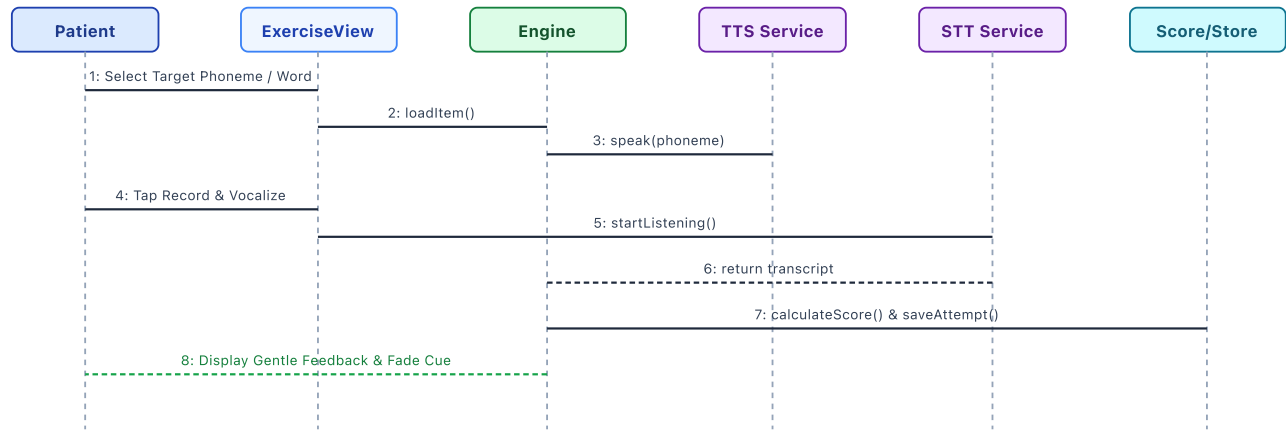
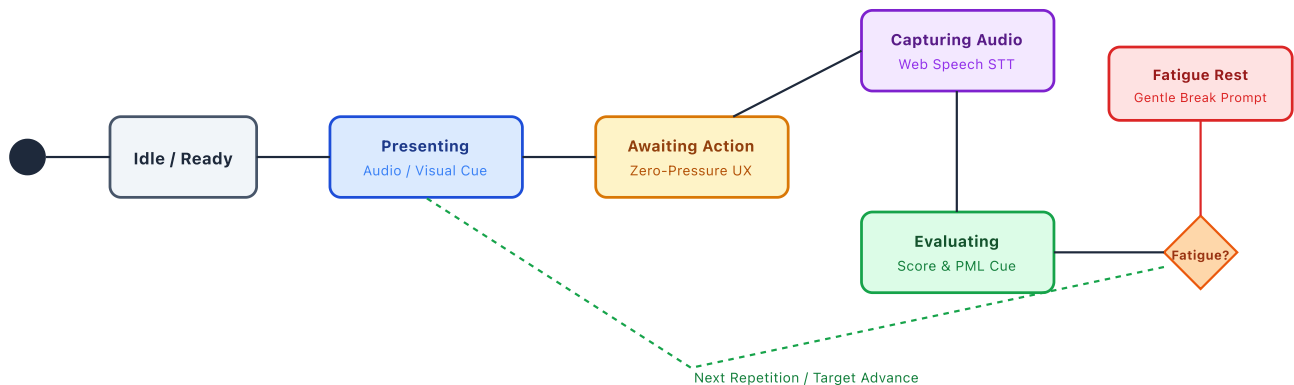


Figure 5: UML State Machine (Exercise Lifecycle & Fatigue Detection) DYNAMIC STATE TRANSITIONS



5. \$50,000 Grant Milestone Allocation

Phase / Milestone	Deliverables	Timeline	Budget
Phase 1: Dysarthric Speech ML	Calibration of client-side lightweight acoustic models for dysphasic speech	Months 1–2	\$15,000
Phase 2: SLP Portal & NDIS Sync	Clinician dashboard, prescription push service, automated NDIS reporting	Months 3–4	\$15,000
Phase 3: Clinical Pilot	6-month longitudinal trial with stroke rehabilitation units in Australia	Months 5–8	\$12,000
Phase 4: Multi-Dialect & Global Access	Phonetic libraries for Australian, UK, and US accents; open-source release	Months 9–10	\$8,000